Antu Saha

Ph.D. Candidate, Department of Computer Science, William & Mary, Williamsburg, VA, USA
asaha02@wm.edu • +1-757-243-3463 • antu-saha.github.io • Google Scholar • GitHub • LinkedIn

RESEARCH INTERESTS

Software Engineering (SE): AI for SE, Agentic SE, Issue Reporting, Issue Resolution, Bug Localization, Software Traceability, Code Rationale Management

EDUCATION

William & Mary, Virginia, USA

Ph.D., Computer Science — Jan 2022 – May 2027 (expected)

- Dissertation Proposal: **Studying and Automating Issue Resolution to Ensure Software Quality**
- Advisor: **Dr. Oscar Chaparro**
- CGPA: 3.96/4.00

Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

B.Sc., Computer Science & Engineering — Mar 2013 – Dec 2017

- **3rd among 120 students** in the class (CGPA: 3.82/4.0)

PUBLICATIONS

Published Papers

[1.] [ICSME'26] A. Adnan*, M. Rahman*, **A. Saha**, and O. Chaparro, "**Rethinking Issue Resolution for AI/ML Systems**," in *Proceedings of the 42nd IEEE International Conference on Software Maintenance and Evolution*, 2026. ***(*M.Sc. students I mentored)*** **[Vision Track]**

[2.] [ICSE'26] **A. Saha**, "**Studying and Automating Issue Resolution for Software Quality**," in *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering*, 2026. **[Doctoral Symposium]**

[3.] [ICSE'25] **A. Saha** and O. Chaparro, "**Decoding the Issue Resolution Process in Practice via Issue Report Analysis: A Case Study of Firefox**," in *Proceedings of the 47th IEEE/ACM International Conference on Software Engineering*, 2025. **[Research Track]**

[4.] [ICPC'25] J. Mahmud*, **A. Saha***, O. Chaparro, K. Moran, and A. Marcus, "**Combining Language and App UI Analysis for the Automated Assessment of Bug Reproduction Steps**," in *Proceedings of the 33rd IEEE/ACM International Conference on Program Comprehension*, 2025. ***(*co-first author: equal contribution)*** **[Research Track]**

[5.] [MSR'25] A. Adnan*, **A. Saha**, and O. Chaparro, "**Sprint: An Assistant for Issue Report Management**," in *Proceedings of the 22nd IEEE/ACM International Conference on Mining Software Repositories*, 2025. ***(*M.Sc. student I mentored)*** **[Tool Demo Track]**

[6.] [ISSTA'24] **A. Saha**, Y. Song, J. Mahmud, Y. Zhou, K. Moran, and O. Chaparro, "**Toward the Automated Localization of Buggy Mobile App UIs from Bug Descriptions**," in *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis*, 2024. **[Research Track]**

[7.] [IJEECS] M. Begum, B. C. Das, M. Z. Hossain, **A. Saha**, and K. A. Papry, "**An Improved Kohonen Self-Organizing Map Clustering Algorithm for High-Dimensional Data Sets**," in *Indonesian Journal of Electrical Engineering and Computer Science*, 2021.

[8.] [ECCE'19] **A. Saha**, J. Saha, and B. Sen, "**An Expert Multi-Modal Person Authentication System Based on Feature Level Fusion of Iris and Retina Recognition**," in *Proceedings of the 2019 International Conference on Electrical, Computer and Communication Engineering*, 2019.

Papers Under Review

[9.] [FSE'27] **A. Saha**, S. Pandey, A. Marcus, K. Moran, and O. Chaparro, "**What Bug Report Information Helps Agents Fix UI-Centric Bugs? An Empirical Study of Android UI Bug Repair**," 2026. **[Research Track]**

[10.] [FSE'27] S. Pandey, J. Wu, J. Mahmud, N. De Silva, **A. Saha**, A. K. Dipongkor, O. Chaparro, M. Fazzini, and K. Moran, "**One Screen, Many Patches: Software Engineering Agents and the Repair of UI-centric Android Bugs**," 2026. **[Research Track]**

[11.] [EMSE] **A. Saha**, M. Sun, and O. Chaparro, "**Evaluating Language Model Applications for Identifying Solution-Related Content in Issue Report Discussions**," *arXiv preprint arXiv:2511.06501*, 2025. ***(Major Revision)***

[12.] [MSR'27] **A. Saha**, A. K. Dipongkor, S. Bennett, K. Moran, A. Marcus, and O. Chaparro, "**Automatically Enhancing the Quality of Android App Bug Reports**," *arXiv preprint arXiv:2604.01148*, 2026. **[Research Track]**

[13.] [TOSEM] M. Sun, **A. Saha**, N. De Silva, A. Mastropaolo, and O. Chaparro, "**Recovering Fine-Grained Code Change Rationale from Multiple Software Artifacts**," *arXiv preprint arXiv:2604.10345*, 2026.

Papers Under Preparation

[14.] [ICPC'27] **A. Saha**, S. Bennett, and O. Chaparro, "**High-Quality Bug Reports from Minimal User Input: An Agentic Approach for Mobile Applications**," 2026. **[Research Track]**

[15.] [ICSE'27] M. Hasan, **A. Saha**, and O. Chaparro, "**SE AI Agent Index: A Dataset of Software Engineering Practices for Agentic AI Systems**," 2026. **[Tool and Data Track]**

[16.] [ICSE'27] M. Hasan, **A. Saha**, and O. Chaparro, "**Trajectiers: A Visualization Tool for AI Coding Agent Trajectories**," 2026. **[Tool and Data Track]**

RESEARCH EXPERIENCE

William & Mary, Virginia, USA

- *Graduate Research Assistant* — Jan 2022 – Present
  - Built **tools and techniques** (AstroBR, BugScribe, Burt++, Sprint) and studied **agentic SE** leveraging **AI coding agents**, **LLMs**, **ML/DL models**, **static/dynamic app analysis**, and **repository mining**
  - Conducted **empirical studies** to construct **large-scale datasets**, **qualitative analysis** of software artifacts, **human studies** (surveys and interviews), and **controlled experiments**
  - Collaborated with researchers from **George Mason University**, **University of Central Florida**, and **University of Dhaka (Bangladesh)**
  - Mentored five Ph.D., two M.Sc., and one undergraduate student in multiple graduate and undergraduate research projects

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- *Lecturer (Instructor of Record)* — Dec 2018 – Jan 2022
  - Collaborated with the faculty research group to improve the **Kohonen self-organizing map** clustering algorithm for **high-dimensional datasets**
  - Supervised three undergraduate students on their theses

Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

- *Undergraduate Student* — Mar 2013 – Dec 2017
  - Designed, implemented, and evaluated iris & retina recognition-based multi-modal person authentication system ***(undergraduate thesis)***

TEACHING EXPERIENCE

William & Mary, Virginia, USA

- *Graduate Teaching Assistant* — Spring 2022–23, Fall 2022
  - Courses: Data Structures, Software Development, Algorithms
  - Prepared assignments and solutions for assignments and exams, conducted office hours, graded coursework, and proctored exams

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- *Lecturer (Instructor of Record)* — Dec 2018 – Jan 2022
  - Courses: Structured Programming, OOP, Information Systems Analysis, Human-Computer Interaction, Numerical Methods, Machine Learning
  - Full course ownership: designed syllabi, conducted theory and lab courses, prepared course materials (*e.g.*, lecture slides, assignments, exams, solutions, *etc.*), proctored exams, graded coursework
  - Supervised students in undergraduate research; student advisor for student well-being

Military Institute of Science and Technology, Dhaka, Bangladesh

- *Lecturer (Instructor of Record)* — Jan 2018 – Dec 2018
  - Courses: Object-Oriented Programming, Structured Programming, Data Structures, Computer Networks, among others
  - Full course ownership: designed syllabi, conducted theory and lab courses, prepared course materials (*e.g.*, lecture slides, assignments, exams, solutions, *etc.*), proctored exams, graded coursework
  - Mentored students in competitive programming

MENTORING

William & Mary, Virginia, USA

- *Graduate Research Mentor* Jan 2022 – Present
  - Mentored five Ph.D. students from the SEA lab, two M.Sc. students and one undergraduate student in various research projects
  - Served as **Graduate International Peer Leader**: supporting new incoming international graduate students at William & Mary

Dhaka University of Engineering & Technology, Dhaka, Bangladesh

- *Undergraduate Thesis Advisor* Dec 2018 – Jan 2022
  - Supervised undergrad students in thesis and competitive programming
  - Advised undergrad students for their mental wellness and academic performance

GRANTS

- **William & Mary Graduate Research Seed Grant'24** for **Decoding the Issue Resolution Process in Practice via Issue Report Analysis: A Case Study of Firefox** research project [ICSE'25]
- **NSF Travel Grant** to attend ICSE'26.
- **SIGSOFT Travel Support (CAPS)** to attend ICSE'26, ISSTA'24.
- **GSA Conference Grant** to attend ICSME'26, ICSE'26, ISSTA'24.
- **WM CS Conference Grant** to attend ICSME'26, ICSE'26, ICSE'25, ISSTA'24.

HONORS & AWARDS

- **International Student Achievement Award** at William & Mary, 2025.
- **Best Poster Award** at W&M CS Symposium for Graduate Studies, 2024 and 2023.
- Technical Education Scholarship, Ministry of Education, Bangladesh, 2013–2017.
- Prime Bank Foundation Scholarship, Prime Bank, Bangladesh, 2013–2017.
- Govt. Merit Scholarship, Ministry of Education, Bangladesh, 2010, 2007, and 2004.

TALKS

Research Paper Presentations

- [ICSME'26] "**Rethinking Issue Resolution for AI/ML Systems**," September 2026, Benevento, Italy
- [ICSE'25] "**Decoding the Issue Resolution Process in Practice via Issue Report Analysis: A Case Study of Firefox**," April 2025, Ottawa, Canada
- [ICPC'25] "**Combining Language and App UI Analysis for the Automated Assessment of Bug Reproduction Steps**," April 2025, Ottawa, Canada
- [MSR'25] "**Sprint: An Assistant for Issue Report Management**," April 2025, Ottawa, Canada
- [ISSTA'24] "**Toward the Automated Localization of Buggy Mobile App UIs from Bug Descriptions**," September 2024, Vienna, Austria
- [ECCE'19] "**An Expert Multi-Modal Person Authentication System Based on Feature Level Fusion of Iris and Retina Recognition**," February 2019, Cox's Bazar, Bangladesh

Poster Presentations

- [ICSE'26] "**Studying and Automating Issue Resolution for Software Quality**," April 2026, Rio de Janeiro, Brazil
- [MSR'25] "**Sprint: An Assistant for Issue Report Management**," April 2025, Ottawa, Canada
- W&M CS Symposium for Graduate Studies, November 2023 & 2024, Virginia, USA

PROFESSIONAL SERVICE

Program Committee Member

- 23rd IEEE/ACM International Conference on Mining Software Repositories (MSR'26)

Conference Session Chair

- 48th IEEE/ACM International Conference on Software Engineering (ICSE'26)

External Reviewer

- IEEE/ACM International Conference on Software Engineering (ICSE'27, '26, '24, '23)
- IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER'27)
- IEEE International Conference on Software Maintenance and Evolution (ICSME'26, '25, '23, '22)
- IEEE/ACM International Conference on Program Comprehension (ICPC'26)
- IEEE International Conference on Technical Debt (TechDebt'26)
- ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE'23)
- Empirical Software Engineering (EMSE)
- ACM Computing Surveys (CSUR)

PROFESSIONAL AFFILIATIONS

- Member, **IEEE**
- Member, **ACM – SIGSOFT**

ORGANIZING & VOLUNTEERING

- Student volunteer at **ICSE'26**, **ICSE'25**, **ICSE'24**, **ESSEC'24**, and **ICSME'23**
- Web chair and student volunteer at **W&M CS Symposium for Graduate Studies** – 2023
- Student volunteer at **W&M CS Symposium for Graduate Studies** – 2022
- Student volunteer at **Grad Bash** – 2023 at **William & Mary**
- Organizer of **DUET CSE FEST'19**, Dhaka University of Engineering & Technology, Bangladesh
- Organizer of **CYBERCOMP'18**, Military Institute of Science and Technology, Bangladesh

REFERENCES

Dr. Oscar Chaparro
Associate Professor
Dept. of Computer Science
William & Mary
Williamsburg, VA 23185, USA
Email: oscarch@wm.edu

Dr. Kevin Moran
Assistant Professor
Dept. of Computer Science
University of Central Florida
Orlando, FL 32816, USA
Email: kpmoran@ucf.edu

Dr. Andrian Marcus
Professor
Dept. of Computer Science
George Mason University
Fairfax, VA 22030, USA
Email: amarcus7@gmu.edu